

MBM University, Jodhpur
Department of Mechanical Engineering
B.E. IInd Year IIIrd Semester
Second Mid-Term Examination Session 2024-25
3ME43A: Kinematics of Machines

Time: 1 hour

Maximum Marks – 10

Q. 1(a). What is Scott-Russel mechanism. What is its limitation? How is it modified? (3)

(b) What is meant by cross-belt drive? Find the length of belt in a cross-belt drive. (3)

Q. 2 (a) The cutting speed of a broaching machine is 12 meter per minute. The cutter of the machine is pulled by a square-threaded screw with a nominal diameter of 72 mm and a pitch 12 mm. The operating nut takes an axial load of 500 N on a flat surface of 90 mm external diameter. Determine the power required to rotate the operating nut. Take $\mu = 0.16$ for all contact surfaces on the nut. (3)

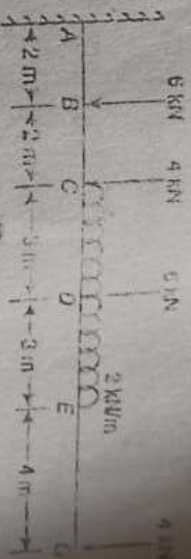
(b) What are various kinds of friction? Discuss each in brief. (1)

TIME: 60 MIN MM:30

MECHANICAL ENGINEERING, MUMBAI UNIVERSITY

CO I Q1. A tension bar tapers from $(D+a)$ diameter to $(D-a)$ diameter. Prove that the error involved in using the mean diameter to calculate the Young's modulus is $(10a/D)^2$ percent. (6)

CO II Q2. Draw the SFD and BMD for the beam shown: (8)



CO II Q3. Derive bending equation for a beam considering pure bending and state the assumptions made. (5)

QUIZ: Write question no. and correct option in answer sheet. (10 = 2x5) (8)

		a.	b.	c.	d.
Q4	If BM is 'M' and 'Z' is section modulus, bending stress is	M/Z	MZ	$M-Z$	$M+Z$
Q5	Figure out odd term	Yield point	Compressive stress	Shear stress	Tensile stress
Q6	Maximum bending moment for a cantilever beam of length 'L' with a udl 'w' is	$wL^2/2$	wL	$wL/2$	wL
Q7	The total extension of a rod of length 'L' and diameter 'D' subjected to a load 'P', is	$4PL/\pi D^2$	$4PL/\pi D$	$PL/\pi D^2$	$PL/4\pi D^2$
Q8	Elongation of a bar of uniform cross section 'A' of length 'L', due to self weight 'W' is given by	$WL/2AE$	$WL/4AE$	$WL/8AE$	WL/AE

MBM University, Jodhpur
Department of Mechanical Engineering
B.E. 11th Year 11th Semester
First Mid-Term Examination Session 2024-25
3ME43A(ME): Kinematics of Machines

Time: 1 hour

Maximum Marks – 10

Q. 1(a). In a slider-crank mechanism, the lengths of the crank and connecting rod are 200 mm and 800 mm respectively; crank has turned 30° . Determine the velocity of slider and the angular acceleration of connecting rod if crank rotates at 40 rad/sec (counter clockwise direction).

(4)

(1)

(b) State Arnold-Kennedy theorem of three centres.

(1)

Q. 2 Define kinematic link, kinematic pair and kinematic chain.

(3)

(b) Sketch and explain the various inversion of double-slider crank chain mechanism.

(1)

(c) Explain Gruebler's criterion for degree of freedom for planar mechanisms.

DEPARTMENT OF MECHANICAL ENGINEERING
MMU UNIVERSITY, JODHPUR

BE II Yr. III SEM.

Material Technology

Paper Code- 3ME42A(ME)

Mid Term Exam.-I

Max. Marks: 10

Date: 25/09/24

All questions compulsory

Q.1	CO1	Describe different crystal structure with diagram.	3
Q.2	CO1	Describe different crystal imperfection with diagram.	3
Q.3	CO1	Copper has F.C.C. structure and the atomic radius $1.278 \text{ \AA}$. Calculate the density of copper crystal. Given atomic weight of copper = 63.5. Avogadro no. = 6.023×10^{23} .	4

II Mid Term Test 2024-25

Class : II Year III Semester (Mechanical Engineering)

Subject : 3ME21A (MA) Mathematics

M.M.: 10

Time : 1hrs.

Do any two questions:

1. (a) Use Bessel formula to find y (25), given that $y(20) = 24, y(24) = 32, y(28) = 35, y(32) = 40$
(b) Evaluate integral limit 0 to 1 $dx/1+x^2$ using Simpson $1/3$ and $3/8$ rule.
2. (a) Apply Gauss – Seidel method to solve $9x - 2y + z = 50, x + 5y - 3z = 18, -2x + 2y + 7z = 19$
(b) Using Runge – Kutta method find an approximate value of y for $x = 0.2$ in steps of 0.1, if $dy/dx = x + y^2$, given $y=1$ when $x=0$
3. (a) Find the Laplace transform of $\sin \sqrt{x}$ Hence show that $L\{(\cos \sqrt{t})/\sqrt{t}\} = (\pi/s)^{1/2} e^{-1/(4s)}$
(b) Prove that the Laplace transform of $\{(\sin^2 t)/t\} = 1/4 \log(s^2 + 4)/s^2$

Engineering Thermodynamics Date 26/09/2024

1st Midterm

Exams Duration: One Hour (11 to 12 PM) M.M.:10

Write general energy equation for a variable flow process.

[1]

2. Establish the equivalence of Kelvin-Planck and Clausius statements.

[2]

3. An inventor comes to an industrialist and claims to have developed a device that operating in a cycle receives 800kJ of heat from a source at 450K and produces 280kJ of net work while rejecting the waste heat to the atmosphere at 300K. What would you advise to the industrialist, he should invest or not? [3]

4. Write the expression of entropy of a closed system.

[2]

5. Write availability functions of closed system and open system.

[2]

Quiz (Maximum Marks:10)

1. Which of the following is the basic of temperature measurement.

[2]

a) Zeroth Law b) First Law c) Second Law d) Third Law

2. All spontaneous processes are

[2]

a) reversible b) irreversible c) impossible d) prone to second law violation

3. Clausius' theorem derives an equation based on the reversible cycle as

[2]

a) $\oint dQ/T = 1^*$ b) $\oint dQ/T = 0$ c) $\oint dQ/T = \infty$ d) $\oint dQ/T \leq 0$

[2]

4. The relationship between first law and second law efficiency is given by

a) $\eta_{II} = \eta_I / \eta_{Carnot}$ b) $\eta_{II} = \eta_I \times \eta_{Carnot}$

In a given heat engine cycle, the relationship between heat supplied (Q), available energy (A.E.) and

ME: 60 MIN MM: 30

Q1. At a point within a body subjected to two mutually perpendicular stresses 120 N/mm^2 and 40 N/mm^2 tensile and 60 N/mm^2 of shear stress. Find σ_1 , σ_n and σ_r on a plane inclined at 45° to normal cross-section. Solve with Mohr's circle method. (8)

Q2. Derive deflection equation. (6)

Q3. A hollow shaft is to transmit 300 KW at 80 rpm . If the shear stress is not to exceed 60 N/mm^2 and $ID = 0.6 OD$, find internal dia and external dia assuming minimum torque is 1.4 times the mean torque. (6)

OR

4. A column $150 \text{ mm} \times 200 \text{ mm}$ is 6 m long, both ends fixed $E = 1.75 \times 10^5 \text{ N/mm}^2$. Find Crippling load and Safe load, if $FoS = 3$. (6)

QUIZ : Write question no. and correct option in answer sheet. (10 = 2x5)

	a.	b.	c.	d.
Q4 Euler formula is applicable for	Short column	Long column	Medium column	Both 'a' and 'b'
Q5 Maximum slope in a simply supported beam with UDL 'w' is	$wL^3/9EI$	$wL^3/16EI$	$wL^3/3EI$	$5wL^3/384EI$
Q6 A column with maximum equivalent length has	both ends hinged	one end fixed and the other end free	one end fixed and other end hinged	both ends fixed
Q7 The load at which a vertical compression member just buckles is known as	Critical load	Crippling load	Buckling load	All of these
Q8 Which of the following statements is true for a simply supported beam?	Deflection at supports is maximum.	Deflection is max. where slope is max.	Slope is minimum at supports	Deflection is max. where slope is zero.

8/10

First Mid Term Exam 2021

Class III & Year III Sem.

Mechanical Engineering Department

Subject: 3ME21A(VIA) : Mathematics

Note: Answer two questions. All questions carry equal marks.

1. Solve $y' = x + 1$ by Monge's Method

(a) $y(1) = 1$, (b) $y(2) = 1$, (c) $y(3) = 1$, (d) $y(4) = 1$

2. A function $y = f(x)$ is defined by $y = x^2 + 2x + 1$

(a) Find the derivative of y with respect to x

(b) Use the Newton's forward difference interpolation formula to compute $y(3.62)$ given $(x, y) = (3, 6)$, $(4, 17)$, $(5, 26)$, $(6, 37)$, $(7, 50)$, $(8, 65)$, $(9, 81)$, $(10, 100)$

(c) Use Lagrange's interpolation formula to find $y(9.5)$ given $(x, y) = (7, 3), (8, 1), (9, 1), (10, 9)$

2/10/21

Page 10

DEPARTMENT OF MECHANICAL ENGINEERING
MBM UNIVERSITY, JODHPUR

MID TERM-II: - 3ME45A(ME): PRODUCTION MACHINE TOOLS

TIME: 1 Hr.

MM: 20

Dt: Nov. 11, 2024

Q1 (CO3): What are the drilling Jigs and Milling Fixtures?

(5)

Q2 (CO4): Explain MIG welding with neat sketch. How does it differ from other metal arc welding processes?

(5)

Q3 (CO5): What are Automatic Machines? List important factors on which the selection of a particular type and size of machining center depends.

(5)

Q4 (CO6): Give the comparison between NC and CNC machines. Discuss briefly 'Adaptive control systems'.

(5)

DEPARTMENT OF MECHANICAL ENGINEERING
MBM UNIVERSITY, JODHPUR
BE II Yr. III SEM.

Material Technology
Paper Code- 3ME42A(ME)
Mid Term Exam.-II

Date: 13/11/24

Max. Marks: 10

All questions compulsory

Q.1	CO2	Describe the role of dislocation in metal and alloys with diagram.	3
Q.2	CO2	Describe effect of different mechanical properties on metal and alloys with diagram.	4
Q.3	CO2	Describe effect of temperature on metal and alloys with diagram	3

A machine is pulled by a rope over a pulley. The operating nut takes an axial load of 500 N on a flat surface of 90 mm external diameter. Determine the power required to rotate the operating nut. Take $\mu = 0.16$ for all contact surfaces on the nut.

(1)

(3)

2nd Midterm

Vidya Today

ME 201A: Engineering Thermodynamics

Date 12/11/2024

Exam Duration: One Hour (12 PM to 1 PM)

Maximum Marks:10

CO5	1. Write combustion equation for fuel $C_{10}H_{22}$	[2]
CO4	2. Write 2 nd Maxwell's equation, first Tds equation and Clausius Clapeyron equation. [3]	
CO3	3. Write the ideal gas equation, Vander-waal gas equation, and equation of Dalton's law of partial pressure. [3]	
CO6	4. Derive the expression for volumetric efficiency of a single stage reciprocating compressor. [2]	